

TSELINA, Russian Federation

WMO: 347470

Lat: 46.550N

Lon: 41.050E

Elev: 112

StdP: 99.99

Time Zone: 3.00 (E03)

Period: 86-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.5	-15.6	-21.0	0.6	-17.9	-18.4	0.7	-15.1	12.1	-9.3	8.7	-0.8	1.4	90	0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	33.2	21.2	31.4	20.6	29.7	19.9	22.5	30.3	21.6	29.0	20.9	27.8	2.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.1	25.3	19.3	14.3	24.5	18.5	13.6	23.8	66.6	30.5	63.5	29.1	60.9	27.9	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.2	5.2	DB	-21.6	35.5	2.8	2.0	-23.7	37.0	-25.3	38.1	-26.9	39.2	-29.0	40.6	(4)
(5)			WB	-21.9	24.3	2.8	1.5	-23.9	25.4	-25.6	26.3	-27.1	27.1	-29.2	28.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-3.2	-3.6	2.5	10.6	16.2	20.6	22.9	21.8	16.0	10.0	2.6	-1.9	(6)	
	DBStd	10.62	5.80	6.01	5.01	4.05	3.64	3.40	2.90	2.90	4.17	4.89	5.59	5.17	(7)	
	HDD10.0	1734	411	381	238	40	3	0	0	0	5	59	227	371	(8)	
	HDD18.3	3569	669	614	492	232	85	16	1	5	93	262	473	628	(9)	
	CDD10.0	1591	0	0	4	59	196	319	399	366	185	58	4	1	(10)	
	CDD18.3	384	0	0	0	1	19	84	142	112	23	2	0	0	(11)	
	CDH23.3	3663	0	0	1	22	190	744	1383	1111	191	21	0	0	(12)	
	CDH26.7	1310	0	0	0	4	36	261	564	399	43	2	0	0	(13)	
(14) Wind	WSAvg	2.1	2.5	2.5	2.8	2.6	1.9	1.6	1.4	1.5	1.6	2.0	2.3	2.4	(14)	
(15) Precipitation	PrecAvg	546	40	36	34	46	50	53	58	52	37	33	46	61	(15)	
	PrecMax	726	98	97	71	137	106	116	190	138	170	80	102	154	(16)	
	PrecMin	333	6	1	2	4	12	0	18	6	0	2	7	6	(17)	
	PrecStd	105	24	26	18	34	27	31	39	45	34	21	27	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	14.0	19.0	25.6	29.7	33.8	36.6	35.6	30.3	25.8	16.9	12.2	(19)	
		MCWB	7.2	8.5	11.6	15.1	18.7	22.1	21.8	22.1	20.2	17.1	13.8	10.4	(20)	
	2%	DB	6.6	8.1	15.5	22.7	27.4	31.6	34.1	32.3	28.0	22.6	13.7	8.5	(21)	
		MCWB	5.5	5.5	9.3	13.8	18.0	20.9	21.7	20.8	19.1	15.4	11.4	7.0	(22)	
	5%	DB	4.9	5.1	12.2	20.0	25.3	29.7	31.8	30.6	25.7	19.6	11.3	5.8	(23)	
		MCWB	3.8	3.5	7.6	12.5	16.8	20.2	21.1	19.9	17.7	14.1	9.8	4.7	(24)	
	10%	DB	3.3	3.0	9.4	18.0	23.2	27.8	29.8	28.8	23.4	17.1	9.5	4.3	(25)	
		MCWB	2.3	2.0	6.2	11.4	16.2	19.9	20.2	19.2	16.9	12.7	8.3	3.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	8.7	11.8	16.6	20.3	23.6	23.8	23.0	21.5	18.0	13.8	10.8	(27)	
		MCDB	8.7	13.7	18.7	23.3	26.9	31.0	31.8	31.7	28.4	24.6	16.4	12.4	(28)	
	2%	WB	5.7	5.7	9.6	14.8	19.0	22.2	22.6	21.8	19.9	16.0	11.8	7.1	(29)	
		MCDB	6.5	7.9	14.4	20.9	24.9	28.9	30.7	29.8	25.3	21.2	13.5	8.3	(30)	
	5%	WB	4.0	3.6	7.9	13.4	17.9	21.3	21.7	21.0	18.6	14.6	10.1	5.0	(31)	
		MCDB	4.6	4.9	11.8	18.7	23.4	27.6	29.5	28.3	24.0	18.6	11.0	5.7	(32)	
	10%	WB	2.3	2.1	6.3	12.2	16.8	20.4	21.0	20.1	17.6	13.3	8.3	3.4	(33)	
		MCDB	3.0	3.0	9.1	16.9	22.1	26.2	28.3	26.9	22.3	16.6	9.4	4.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.6	7.3	10.1	10.4	10.1	11.2	11.9	10.8	9.4	5.8	5.1	(35)	
		MCDBR	5.8	8.3	12.3	14.1	13.5	13.1	14.3	14.4	13.5	12.8	8.4	6.1	(36)	
	5% WB	MCWBR	4.7	6.0	7.2	7.2	6.2	5.2	5.3	6.0	6.5	6.8	6.0	5.1	(37)	
		MCWBR	5.6	8.0	11.9	12.6	11.4	11.2	12.2	12.3	11.8	11.5	7.8	6.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.341	0.375	0.413	0.453	0.419	0.425	0.450	0.448	0.418	0.386	0.354	0.346	(40)	
		taud	2.229	2.141	2.124	2.064	2.250	2.265	2.203	2.202	2.254	2.298	2.318	2.242	(41)	
	Edn at Noon	Edn at Noon	720	768	798	802	849	844	815	799	788	754	706	664	(42)	
		Edh at Noon	84	112	133	154	133	131	138	133	116	96	77	75	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.06	1.92	2.95	4.64	5.95	6.40	6.36	5.73	4.18	2.49	1.25	0.84	(44)	
	RadStd	RadStd	0.22	0.33	0.27	0.45	0.62	0.46	0.32	0.41	0.41	0.32	0.20	0.10	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (7 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	+127	+84	(47)

Nomenclature: See separate page